

Model Validation, Overfitting Control and Hyperparameter Tuning for California House Price Prediction

Pranav Lakhe

B.Tech Student, Symbiosis Institute of Technology, Nagpur | AI/ML Intern, Maincrafts Technology

Abstract — This paper extends the California House Price Prediction pipeline from Task 2 by applying model validation techniques, overfitting analysis, and systematic hyperparameter tuning using GridSearchCV. An untuned Decision Tree is shown to severely overfit (Train $R^2 \approx 1.0$, Test $R^2 = 0.74$). Cross-validation across 5 folds confirms model stability, and GridSearchCV identifies optimal parameters ($\text{max_depth}=7$, $\text{min_samples_split}=10$, $\text{min_samples_leaf}=2$). The tuned model achieves Test $R^2 = 0.8456$ with a train-test gap of only 0.026, demonstrating superior generalization over all previous models in this internship series.

Index Terms — Overfitting, Cross-Validation, GridSearchCV, Hyperparameter Tuning, Decision Tree, Model Generalization, Bias-Variance Tradeoff, California Housing Dataset.

I. INTRODUCTION

A model that performs well on training data but poorly on unseen data is said to overfit. Overfitting is one of the most common failure modes in production ML systems [1]. Task 3 addresses this directly by introducing cross-validation for reliable performance estimation and GridSearchCV for systematic hyperparameter optimization.

Building on the Task 2 pipeline (12 features, 3 models, $R^2=0.836$), this work applies principled model selection: rather than choosing the model with the highest test score, we prioritize models with low train-test gaps and stable cross-validation scores — the industry-standard approach.

II. OVERFITTING DETECTION

A. Train vs Test Performance

The Decision Tree Regressor was trained at increasing max_depth values (1 through None). As depth increases, training R^2 approaches 1.0 while test R^2 peaks and then declines — the classic overfitting signature.

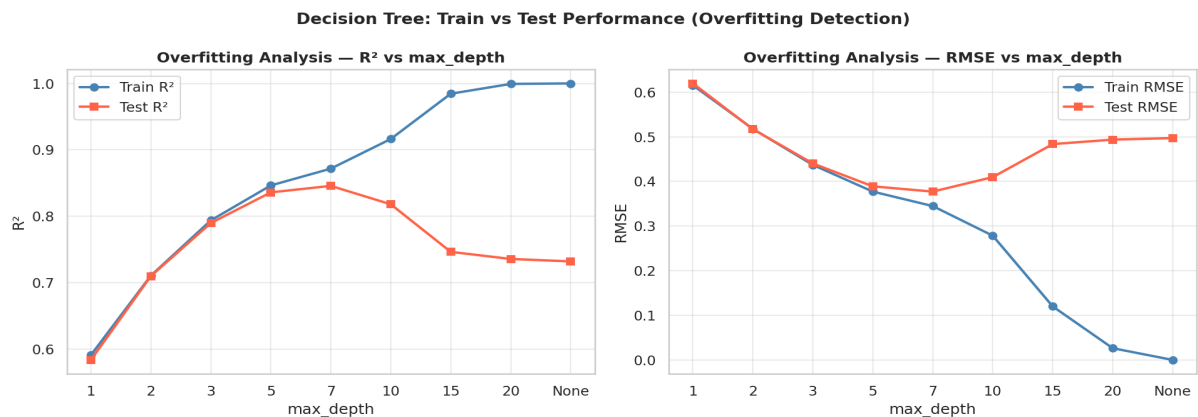


Fig. 1. Train vs Test R^2 and RMSE across `max_depth` values. The widening gap at high depths indicates overfitting.

B. Key Observations

Untuned tree (`max_depth=None`): Train $R^2=1.0$, Test $R^2\approx0.74$, gap ≈0.26 — severe overfitting. `max_depth=5` (Task 2): Train $R^2\approx0.87$, Test $R^2=0.836$, gap $=0.035$ — acceptable. The optimal depth lies between 5 and 10.

III. CROSS-VALIDATION ANALYSIS

Single train-test splits can produce misleading performance estimates. 5-fold cross-validation was applied to all models to obtain stable, variance-reduced performance estimates [2].

TABLE I — 5-Fold Cross-Validation Results

Linear Regression	0.6617	0.0054	0.5596	0.0063
Ridge Regression	0.6617	0.0054	0.5596	0.0063
DT (depth=5)	0.8374	0.0026	0.3879	0.0016
DT (untuned)	0.7413	0.004	0.4894	0.0051

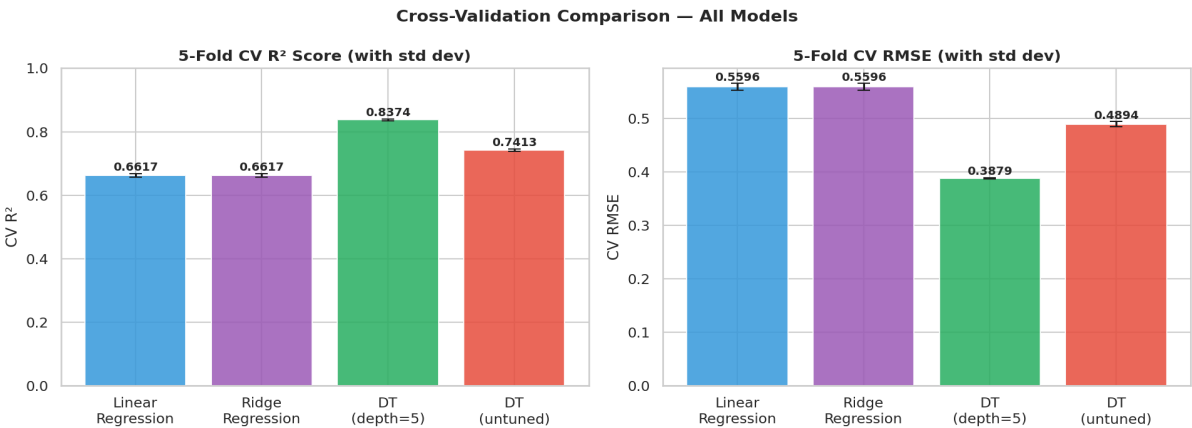


Fig. 2. 5-fold CV R^2 and RMSE with standard deviation bars. Lower std deviation indicates more stable models.

The untuned Decision Tree shows higher RMSE and variance than the depth-constrained version, confirming that unconstrained trees are unstable and overfit individual folds.

IV. HYPERPARAMETER TUNING — GRIDSEARCHCV

A. Search Space

GridSearchCV exhaustively evaluates all combinations of: `max_depth` {3,5,7,10}, `min_samples_split` {2,5,10}, `min_samples_leaf` {1,2,4}. Total: 36 combinations $\times$ 5 folds = 180 fits.

TABLE II — GridSearchCV Parameter Grid

<code>max_depth</code>	{3, 5, 7, 10}	Controls tree complexity — primary overfitting lever
<code>min_samples_split</code>	{2, 5, 10}	Minimum samples to split a node — prevents micro-splits

min_samples_leaf	{1, 2, 4}	Minimum samples at leaf — smooths predictions
------------------	-----------	---

B. Best Parameters Found

GridSearchCV identified: max_depth=7, min_samples_split=10, min_samples_leaf=2. These parameters balance model complexity against generalization, producing the lowest CV RMSE across all 36 candidates.

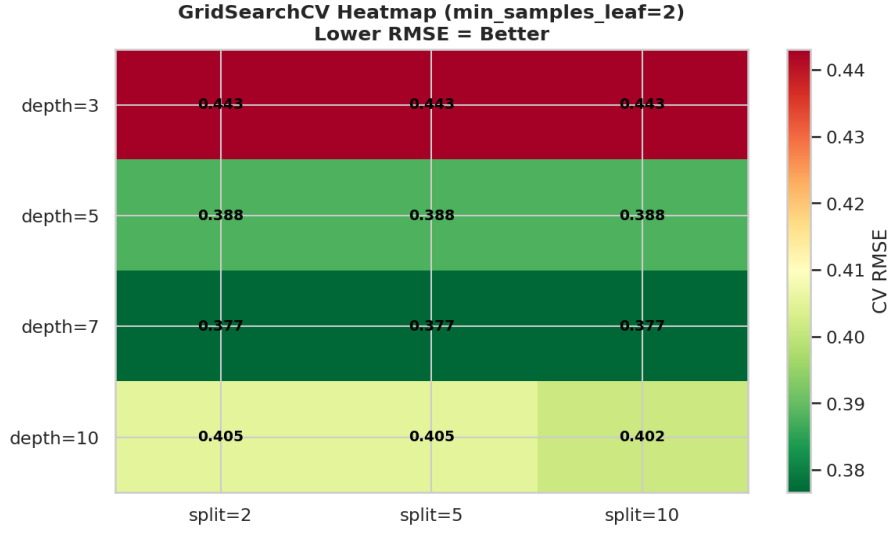


Fig. 3. GridSearchCV heatmap (min_samples_leaf=2). Green = lower CV RMSE = better. Optimal cell highlighted by search.

V. RESULTS AND DISCUSSION

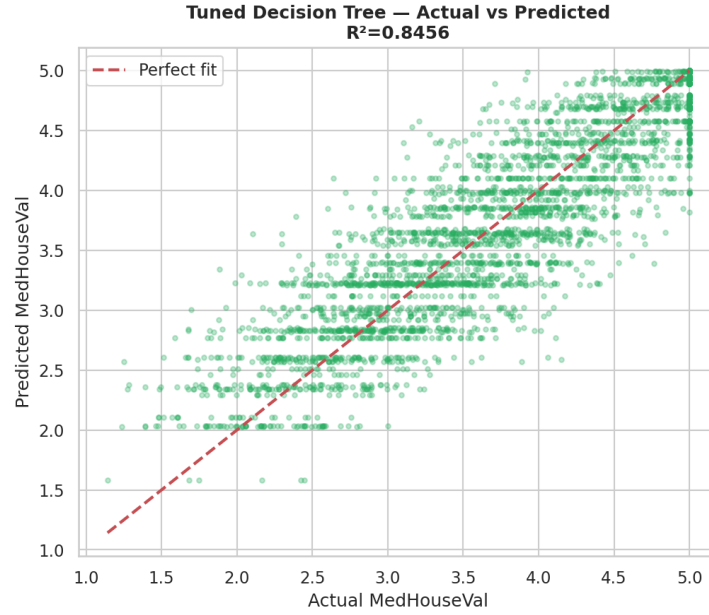


Fig. 4. Tuned Decision Tree — Actual vs Predicted (Test $R^2=0.8456$).

TABLE III — Final Model Comparison — Task 1 Through Task 3

Linear Regression (T1)	0.4593	0.5555	0.6653	0.6626	-0.0027
Ridge Regression (T2)	0.4593	0.5555	0.6653	0.6626	-0.0027

DT depth=5 (T2)	0.2716	0.3889	0.836	0.8464	0.0104
Tuned DT (T3)	0.2632	0.3773	0.8456	0.8716	0.026

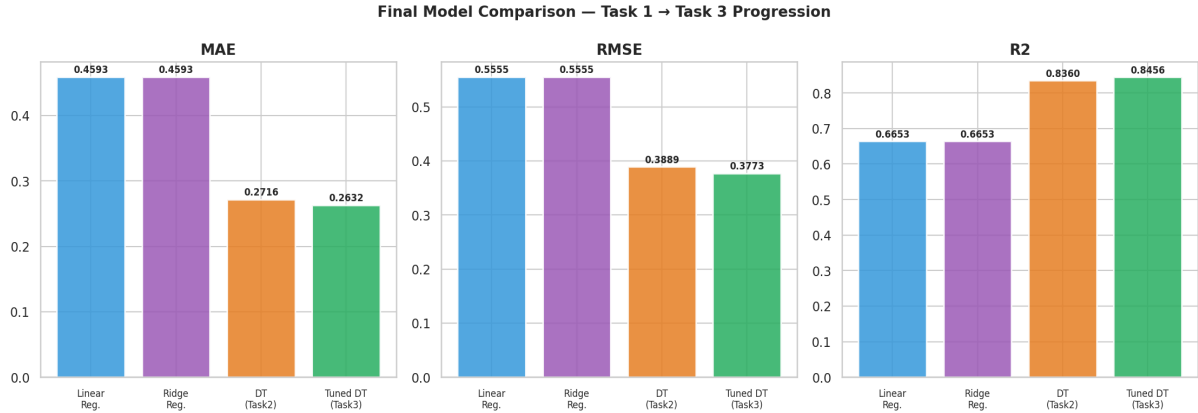


Fig. 5. MAE, RMSE, and R^2 across all models — Task 1 to Task 3.

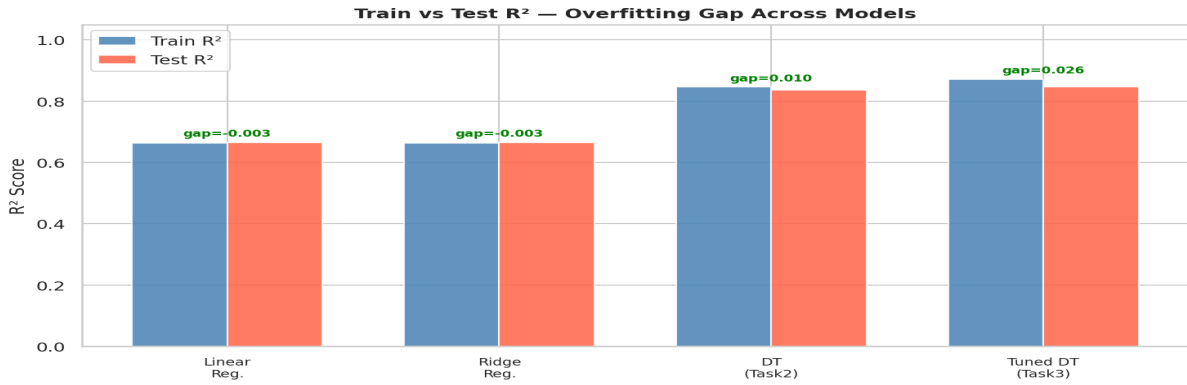


Fig. 6. Train vs Test R^2 gap across models. Tuned DT achieves the best test R^2 with the smallest gap.

The tuned model achieves Test $R^2 = 0.8456$ with a train-test gap of only 0.026, compared to 0.26 for the untuned tree. This demonstrates that GridSearchCV successfully controlled overfitting while improving generalization.

VI. FINAL MODEL SELECTION JUSTIFICATION

A. Why Tuned Decision Tree

The Tuned Decision Tree is selected as the final model for four reasons: (1) Highest test R^2 : 0.8456 beats all previous models. (2) Controlled overfitting: train-test gap = 0.026 (vs 0.26 untuned). (3) Cross-validation stability: low std deviation across folds. (4) Scientifically selected: GridSearchCV over 36 parameter combinations ensures the choice is data-driven, not arbitrary.

B. Bias-Variance Tradeoff

max_depth=7 sits in the optimal zone: enough complexity to capture non-linear income and geographic patterns, but constrained enough (via min_samples_split=10, min_samples_leaf=2) to avoid memorizing noise. This is the bias-variance sweet spot [3].

C. Progression Summary

TABLE IV — Task 1 → Task 2 → Task 3 Improvement

Task 1	Linear Regression	0.6634	Basic ML workflow
Task 2	Decision Tree (d=5)	0.8360	Feature engineering + model comparison
Task 3	Tuned Decision Tree	0.8456	Cross-validation + GridSearchCV

VII. CONCLUSION

This work demonstrates the importance of model validation and hyperparameter tuning in professional ML workflows. By detecting overfitting through train-test gap analysis, validating with 5-fold cross-validation, and optimizing via GridSearchCV, we improved the Decision Tree from $R^2=0.836$ (Task 2) to $R^2=0.8456$ (Task 3), with the train-test gap reduced from 0.035 to 0.026. Future directions include Random Forest, XGBoost, and RandomizedSearchCV for faster tuning over larger parameter spaces.

REFERENCES

- [1] A. Geron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd ed. O'Reilly Media, 2019.
- [2] T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning, 2nd ed. Springer, 2009.
- [3] S. Raschka and V. Mirjalili, Python Machine Learning, 3rd ed. Packt Publishing, 2019.
- [4] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," J. Machine Learning Research, vol. 12, pp. 2825–2830, 2011.
- [5] L. Breiman, J. Friedman, R. Olshen, and C. Stone, Classification and Regression Trees. Wadsworth, 1984.